Mayank Shivhare

Senior Test & AI Automation Engineer | ISTQB Certified

Singapore | +65 86176670 | mayank.shivhare@gmail.com

LinkedIn: linkedin.com/in/mayank-shivhare

Summary

ISTQB-certified test engineer with **11+** years of experience across automotive and industrial automation - Wipro, Capgemini, Continental, and Rockwell Automation. Specialised **in firmware validation, HIL testing, and infotainment systems** for OEMs including Honda, Toyota, Nissan, and Audi. Built Python and Jenkins-based automation frameworks that cut testing cycles by 30–35% and achieved 95%+ requirement coverage on safety-critical programmes. At Rockwell, I **extended this into AI —** developing Microsoft Copilot and GitHub Copilot **agents** that reduced test case, script creation and review effort by up to 75%. Experienced in ISO 26262, ASPICE, and cross-functional delivery across Singapore, India, and on-site OEM environments.

Tools and Skills

AI & Automation: GitHub Copilot (Custom Agents Development), Microsoft Copilot (Custom Agents), Model Context Protocol, Multi-Agentic Workflows, LLM Engineering, RAG, Test Automation Framework Design, CI/CD (Jenkins)

Testing & Validation: Firmware Validation, HIL Testing, System Integration Testing, Black-Box Testing, Functional Testing, Regression Testing, Requirement Analysis, Test Planning, Defect Life Cycle, STLC

Tools & Platforms: Python, Jenkins, CANoe, dSpace, ControlDesk, AutoTest, MxSuite, Insight Explorer, Studio 5000, ControlFlash, K2L Mocca, K2L Viewer, KeySight, PyCharm, Wireshark, TortoiseSVN, Git

Test Management & Collaboration: qTest, Jira, Jama, IBM DOORS, Confluence, Code Collaborator

Protocols & Standards: CAN, LIN, MOST, Ethernet, CXPI, ISO 26262, ASPICE

Certifications

- Certified Tester Foundation Level (CTFL): ISTQB Certified (via ASTQB), **ISTQB** Certified in the US
- **Google DeepMind**: AI Research Foundations - In progress.
- **AI Engineer Core Track**: LLM Engineering, RAG, QLoRA, Agents - In progress.
- **Agile** Software Development from the University of Minnesota
- Introduction to **Artificial Intelligence** from IBM

Work History

Rockwell Automation, Singapore

Senior Functional Test Engineer | Feb 2023 – Present

503X I/O Modules - Analog, Digital & Safety firmware validation

- Developed and executed **comprehensive test plans** for **503X I/O Modules** (Analog, Digital, Safety).
- Performed firmware validation and root-cause analysis across analog, digital, and safety I/O modules, **preventing production escapes and improving firmware stability across releases**.
- Led cross-functional defect triage and design reviews with hardware and firmware teams, **reducing rework cycles and accelerating issue closure**.
- Conducted **firmware flashing** of module base, adapter, and module firmware using Control Flash, ensuring accurate and timely updates.
- Validated a wide range of analog and safety-critical features including No-Load Detection, Short-Circuit Detection, Calibration, Scan Time, and alarm handling across multiple firmware releases.
- Automated test cases using **Python**, reducing execution time by **25%**.
- Identified, debugged, and tracked firmware and test-environment defects in Jira, **driving timely fixes and maintaining zero critical escape rate**.
- Maintained **100%** requirement traceability by mapping test cases (qTest) to functional requirements (JAMA).

AI & Copilot Initiatives:

- Architected end-to-end AI integration using **multi-agentic workflows** across the full testing lifecycle—spanning autonomous test case creation from FRS/FRT (**qTest**), AI-led peer reviews (**Code Collaborator**), Python script generation, and autonomous **Jira** anomaly raising—leveraging **Microsoft Copilot custom agents with MCP**.
- Cut test case creation and review effort by ~50% by training a Microsoft Copilot agent on FRS/FRT documentation and historical test cases to autonomously generate test cases in qTest and review them in Code Collaborator.
- Slashed test script creation and review effort by ~75% by building a GitHub Copilot agent that auto-generates Python test scripts referencing qTest test cases, enforces Rockwell architecture/coding standards, injects all required supporting functions, and posts inline review comments directly in Code Collaborator.
- Engineered an AI-powered review agent that cross-references Python test scripts against qTest test cases to surface discrepancies and flag potential defects — eliminating manual script audits.
- Developed an internal AI tool to automate qTest property management, commenting, and Jira anomaly creation, reducing repetitive overhead and enabling autonomous defect triage.
- Engineered a domain-specific RAG (Retrieval-Augmented Generation) pipeline to answer internal domain questions with high accuracy, using ChromaDB as the vector store and open-source LLMs (Qwen via Ollama) for inference. Implemented hybrid retrieval combining vector search and BM25 keyword search, topped with a reranker to deliver precision-optimised, context-aware answers over internal documentation.

Continental Automotive, Singapore

Senior Embedded System Engineer| Jan 2022 – Feb 2023

Nissan, Audi, Suzuki: Automated and Manual Validation of Instrument Panel Cluster and Infotainment.

Team Size: 7 (Lead-1, Tester-5, Manager-1)

- Led **Automated & Manual Validation** for Nissan, Audi, and Suzuki IPC & Infotainment Systems.
- Authored and executed specification-driven test cases using Python-based uTASS automation**, improving test repeatability and execution efficiency.**
- Validated **Intelligent Traffic Systems (ITS)** features like **Lane Departure Warning, Blind Spot Detection, Park Assist, and Tire Pressure Monitoring System (TPMS)**.
- Spearheaded **bug tracking and resolution**, reporting over **100+ critical defects** using **JIRA (highest in the team)**.
- Maintained **requirement traceability** by mapping test cases to specifications in **DOORS**.
- Collaborated with cross-functional teams, including developers and UI/UX designers, to optimize infotainment system functionality.
- Flashed **Application (AC)** and **Graphics (GC)** software for clusters, ensuring compatibility with regional configurations

Capgemini

Senior HIL Test Engineer | Jun 2019- Dec 2021

Honda & Toyota IPC - HIL and Infotainment Testing (dSpace, CANoe)

Team Size: 12 (Lead-1, Tester-10, Manager-1)

- Authored HIL automation scripts for 13 instrument cluster modules (Speedometer, Fuel, Navigation, HMI, CXPI, and more) using DENSO AutoTest and dSpace, increasing automated test coverage from 40% to 80% on Honda and Toyota platforms
- Executed **manual** verification of test cases of the above-mentioned modules using **Autotest, CANoe, KeySight, and K2L Viewer**.
- Executed HIL validation for Honda and Toyota IPC using dSpace + ControlDesk, running 500+ automated test cases per cycle and reducing hardware-in-the-loop test execution time by ~30% through script optimisation.
- **GDC and VM** Flashing of hardware using the Renesas tool.
- Conducted multimedia testing on meters using CAN and MOST protocols, ensuring optimal performance.
- Performed bidirectional protocol conversion between CXPI, CAN, and MOST, enhancing system compatibility.
- Led a proof-of-concept with Danlaw MxSuite to automate 30+ test scenarios, evaluating its viability as a replacement automation framework for the HIL environment.
- Conducted change point and regression analysis across 8 firmware versions for Honda IPC, identifying 80+ requirement gaps early and preventing post-release defect escapes through early-stage gap closure.

Wipro Limited

Senior Software Test Engineer | Jan 2015 – Jun 2019 | 4 years 5 months

Honda Infotainment - Automation, CI/CD Framework & Manual Validation.

Team Size*:* 20 (Lead-3, Developer-17)

- Developed Python automation scripts covering 12+ infotainment modules (CarPlay, Android Auto, Voice Recognition, OTA, Navigation, and more), reducing manual regression cycle time by ~35% and enabling nightly test runs on the Honda platform.
- Led a 6-engineer test team across 4 delivery milestones, ensuring on-time handoffs with zero critical escapes reported by the OEM.
- Architected a Jenkins-based CI/CD pipeline that automated test execution and report generation for a 20-member team, cutting release validation time from 5 days to 2 days and eliminating manual reporting overhead.
- Achieved 95%+ test coverage across Honda infotainment modules through combined manual and automated validation, ensuring compliance with OEM acceptance standards across major software releases.
- Collaborated with on-site teams for seamless communication and project delivery.

Education

- Lakshmi Narain College of Technology, Bhopal-462021 (2014)
- Bachelor of Engineering - BE, Information Technology, 7.35

Honors & Awards

- **Employee Excellence Award** (2018): Led team to deliver 30% faster testing cycles.
- **Shining Star Award** (2017): Recognized for automating critical Infotainment test scripts.
- **PES Excellence Award** (2016): Streamlined defect management process